Simulating Reddit Echo Chamber Communities Through Representative LLM Persona Agents

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Abstract

We create multiple LLM-based agents which simulate the real world thoughts and opinions of identified echo chamber communities online. Our motivation stems from recent escalations of polarized ideologies that have manifested into real world settings. Personalization algorithms have been instrumental in the entrenchment of beliefs that complicate mutual understanding in many social settings. In our project we leverage LLM advancements and employ topic modeling and sentiment analysis to detect echo chamber communities online, introduce a novel way of testing these agents on large scale real-world survey data, and predict the trajectory of these communities on novel topics before they transpire online. The first part of this paper details the method to identify echo chamber communities on crawled Reddit data; the second part details the development of LLM Persona agents and testing against real world survey data. We also carry out a multiagent discussion simulation to extract insights on new topics.

Keywords: persona agents, LLM agents, social simulation, polarization, echo chamber

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Echo chambers and their resulting communities have a significant impact on societal dynamics (Couldry, 2020). Recent events have highlighted the concerning trends within echo chamber communities, including the rapid spread of misinformation surrounding events like the COVID-19 pandemic and political elections, the increasing entrenchment of polarized views, and the potential for online ideological radicalization to manifest into real-world brutality (O'Connor & Weatherall, 2019). The role of social media algorithms in inadvertently amplifying these dynamics and the resulting fragmentation of shared reality further emphasize the urgency of gaining a deeper understanding of these phenomena. The difficulty in predicting the trajectory and impact of these communities necessitates innovative approaches to analysis and modeling.

Echo Chambers

An echo chamber is often understood as an environment where individuals are repeatedly exposed to information, opinions and interaction that reinforce their existing beliefs, while the opposing viewpoints are filtered out or marginalized. As Sunstein (2001) noted, it has become central to discussions of online communication and such an environment provides repeated affirmation of shared views, intensifying ideological polarization, and causing mitigation in the opportunities presented by cross-cutting discourses.

Nowadays, the rise of social media platforms has brought specific concern about echo chambers, especially due to their algorithmic content curation, selective exposure, and homophilous social network. All these factors will encourage users to engage mainly with their ideologically aligned content and communities (Pariser, 2011). Due to this reason, online platforms are often characterized as spaces where political polarization is exaggerated rather than a platform helping discussion across ideological divides. However, the prevalence and strength of echo chambers remains mixed within several empirical studies.

Some studies do find strong patterns of echo chambers (Conover et al., 2011), while others find indications that users are not completely insulated from cross-ideological content and may even frequently engage with out-group perspectives (Garrett, 2009).

Emphasizing the interaction patterns of in-group and out-group alone does not provide a complete account of the echo chamber dynamic, as it reflects disagreement or conflict rather than ideological openness and does not necessarily weaken belief reinforcement (Bail et al., 2018). Our project takes a multidimensional approach by primarily focusing on sources of information, expressed ideological stance, and the historical interaction between subreddits. In this respect, in-group versus out-group interactions are supplementary analysis. This is so because such focus can further allow nuanced exploration of how an ideological environment will be manifested over time through certain discourses

LLM Agents as Demographic Representations

LLM (Large Language Model) social simulations have been identified as a promising avenue towards social understanding and supporting future research. As a relatively new advent following the success of machine learning and notably LLMs, social simulations can aid exploratory social research such as pilot studies and provide insights into the requirements for an AI system to be more humanlike (Anthis et al., 2025). From a social science perspective, this project offers a valuable avenue for exploring critical phenomena such as political, social and ideological polarization (Haque et al., 2023), the diffusion of information and misinformation alike (Zheng & Tang, 2024), the formation of group identity (Törnberg, 2018), and the potential for predicting behavioural patterns within these insular online environments (Lorig et al., 2021).

Role-Playing Language Agents (RPLAs) are a type of Natural Language Persona Agents (NLPAs) that simulate a specified role, largely of human nature. Demographic personas are a subset of RPLAs which represent a group of individuals through their shared demographic qualities. Common characteristics may

be occupations, ethnic groups, and personality types (Chen et al., 2024). These personas are inherent in LLMs, and role-playing them exploits the statistical stereotypes in LLMs, as well as the linguistic style, professional knowledge, and behavioral nuances representative of a demographic archetype. Interestingly, from a political science standpoint, Argyle et al. (2023) reinforces this idea, stating that the biases these machines perpetuate could actually be of good use by providing proxy human participants in significant numbers, a known challenge in social science research. In their study, they tested GPT-3's "algorithmic fidelity" – the degree to which the complex patterns of relationships between ideas, attitudes, and sociocultural contexts within a model accurately mirror those within a range of human subpopulations – confirming that with appropriate context, these demographic persona agents respond similarly to their human counterparts politically. This further reinforces our motivation of developing LLM agents for social science insights.

Social Simulations using LLM Agents

Beyond analysis and agent building, we tested our agents on novel topics by simulating real world debates between multiple refined demographic persona agents. RPLAs have demonstrated remarkable capabilities in simulating nuanced, human-like interactions across various environments. Furthermore, recent findings in multi-agent interaction environments have revealed that diversifying the types of agents, scaling up their number, and increasing interactions, lead to the emergence of unplanned social behaviors. Such behaviors arise spontaneously from discussions among multiple agents, highlighting the potential for complex, dynamic systems within LLM architectures. Zhang et al. (2024) also showed that human level phenomena can be replicated and provide future insights in collaborative environments. They simulated collaborative environments between agents programmed with certain traits ("*easygoing*" vs "*confident*") and in sequences of debates and reflections. A key difference between this experiment and our intended direction lies in the final output; the agents in their study had to complete a task instead

of debate on opinions where a consensus may not need to be reached. Nevertheless, we believe that such a simulation could reveal insights that leverage the abundance of echo chamber communities online and their relationship to real world social consequences.

Contribution of present study

The insights gained from these agents can contribute to the development of more effective intervention strategies by policymakers and social media platforms aimed at fostering dialogue and critical thinking, narrowing the degree of polarization. The nature of echo chambers themselves presents difficulties in the participating communities to interact outside of their own bubble. The simulation may make it possible to envision scenarios in which they do, and find a way to neutrality. Educators and civil society organizations can use the knowledge to develop media literacy initiatives and promote constructive dialogue. By shedding light on the internal workings of echo chambers, this research ultimately seeks to empower individuals to navigate the digital landscape more critically and contribute to a more informed and less polarized society.

Method

Data Collection

Data was collected from Reddit using the official Reddit API through Python Reddit API Wrapper (PRAW). The dataset consists of posts, comments and users' data from a set of target political subreddits (r/Conservative and r/Liberal).

For each target subreddit, posts were retrieved from two categories (Monthly Top Posts and Hot Posts) to balance out popularity-based and visibility-based contents. These two streams were merged into one and then deduplicated by post ID to avoid counting the same submission twice when it appeared in both lists. Each unique submission was then categorized into a post type using text/URL heuristics, where text was non-empty selftext and links were external URLs not containing 'reddit.com'. Table 1 shows the amount retrieved from the process.

Posts				
Subreddit	Text posts (n)	Link posts (n)	Total posts (N)	% of posts
r/Conservative	64	575	639	56.1
r/Liberal	265	235	500	43.9
Total	**329**	**810**	**1139**	**100**

Table 1 *Number distribution of posts retrieved from Reddit*

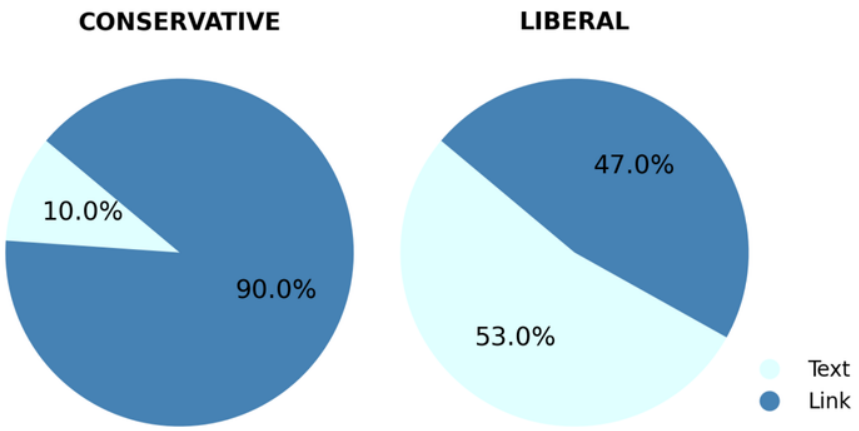

Figure 1 *Proportion of texts and links in each r/Conservative (left) and r/Liberal (right)*

For every post collected, comments were pulled out by first loading a full comment tree and then flattening the thread into a list for sampling. Because highly upvoted comments tend to dominate analyses, we intentionally sampled comments to capture multiple levels of visibility per post. Specifically, for each post it selected up to 150 comments in total, split evenly as shown in Table 2.

Comment Sampling Parameters per Post		
Sampling component	Target (n)	Operational definition
Top comments	50	Highest-score comments
Mid-range comments	50	Comments around the median score rank
Random comments	50	Random sample from all available comments in the post
Total target per post	**150**	**Combined then deduplicated by comment ID**

Table 2 *Operational definitions for each sampling component of a retrieved post*

These three collections were then combined and deduplicated by comment ID, which means the final count of comments gathered would be less if threads were small, comments were deleted/removed by Reddit, or duplicates existed across the three sampling groups.

Comments		
Subreddit	Comments collected (n)	% of comments
r/Conservative	13,664	55.8
r/Liberal	10,823	44.2
Total	**24,487**	**100.0**

Table 3 *Final number and percentage of all comments collected after preprocessing*

To capture users' broader participation patterns by analyzing prior community exposure, we gathered a set of unique usernames that appeared as post authors and comment authors, and their respective past subreddits history, including a set of unique subreddits each user commented in and the comment frequency in each subreddit.

Unique Users Observed by Subreddit		
Subreddit	Users collected	% of users
r/Conservative	2,493	39.4
r/Liberal	3,837	60.6
Random comments	**6,330**	**100.0**

Table 4 *Number and percentage of unique users observed by subreddit*

Bias Labelling and Topic Modelling

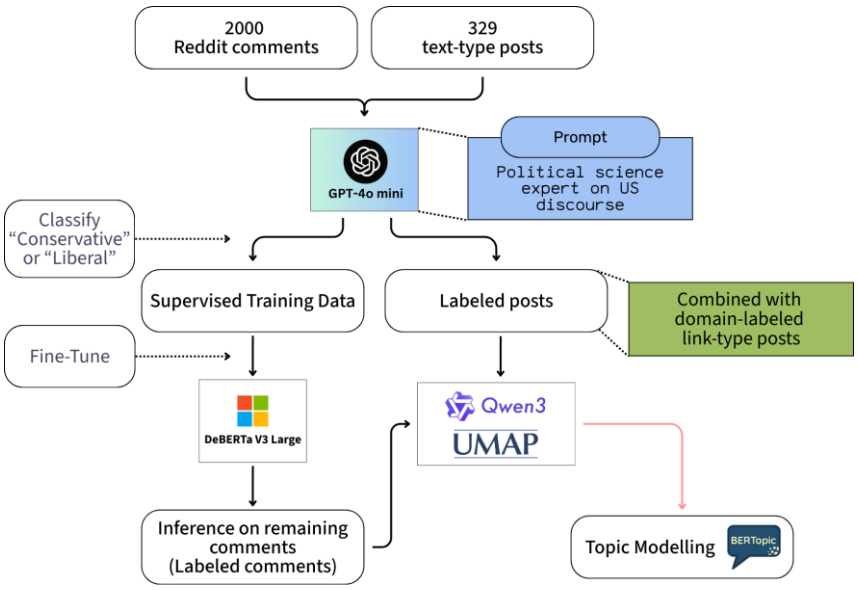

Figure 2 *Bias labelling and Topic modelling pipeline*

To incorporate source-level analysis, link-type posts were assigned domain-level bias labels using the AllSides media bias ratings. For each link-type post, the domain of the external link was extracted and cross-referenced with the AllSides database and when a match was found, the corresponding bias rating was used to label the post. For domains not listed in AllSides, manual labeling was conducted, with particular attention given to content from x.com, where links are often associated with specific user accounts (e.g., x.com/author). These posts were incorporated into the dataset as the domain-labeled link-type component.

The labelling process followed the workflow illustrated in Figure 2. The process began with a curated sample of evenly split 2,000 Reddit comments and all 329 text-type posts. These data were fed to GPT-4o mini, prompted under the persona of a "political science expert on U.S. discourse." Under this instruction, the model was tasked with labeling each entry as either "*Conservative*" or "*Liberal*." These outputs served as supervised training data and also provided a labeled subset of text-type posts.

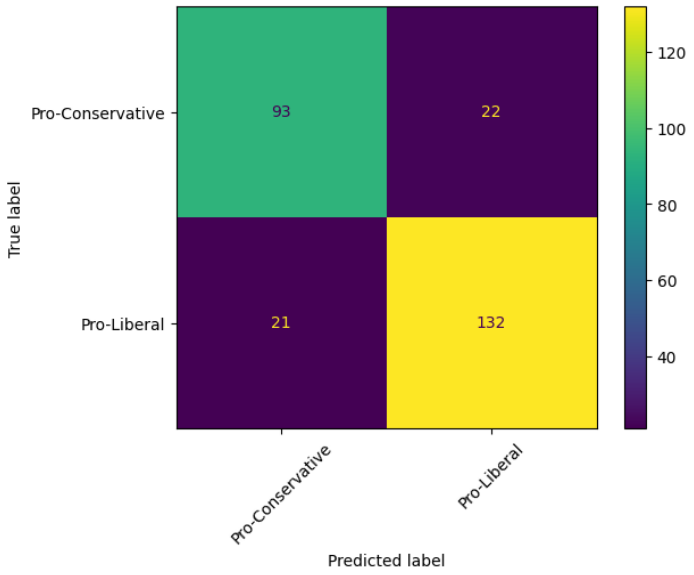

Figure 3 *Confusion matrix of fine-tuned DeBERTA v3 Large*

This labeled comment data was then used to fine-tune Microsoft's DeBERTa V3 Large model, which was subsequently applied to classify the other remaining unlabeled Reddit comments. At the same time, the labeled text posts were merged with the domain-labeled link-type posts, producing a combined dataset that includes ideological stance based on domain used (link-type posts) and LLM labeling (text-type posts).

Finally, the combined set of labeled posts and comments were combined, embedded using Qwen3-Embedding-8B and projected into a lower-dimensional space using UMAP to facilitate structural analysis. The resulting representations were then used as input for topic modeling with BERTopic.

LLM Persona Agent Development

Using the frequent topics identified in the previous stage, we constructed distinct LLM personas based on the topics as a general representation and their representative comments to guide the refinement of such personas. We used Google's newest release of their Gemini language model, Gemini

3 Pro, to create these personas. Google has been at the forefront of LLM advancements, boasting highly sophisticated reasoning capabilities in their models. We chose to use their model for this very reason as well as its accessibility.

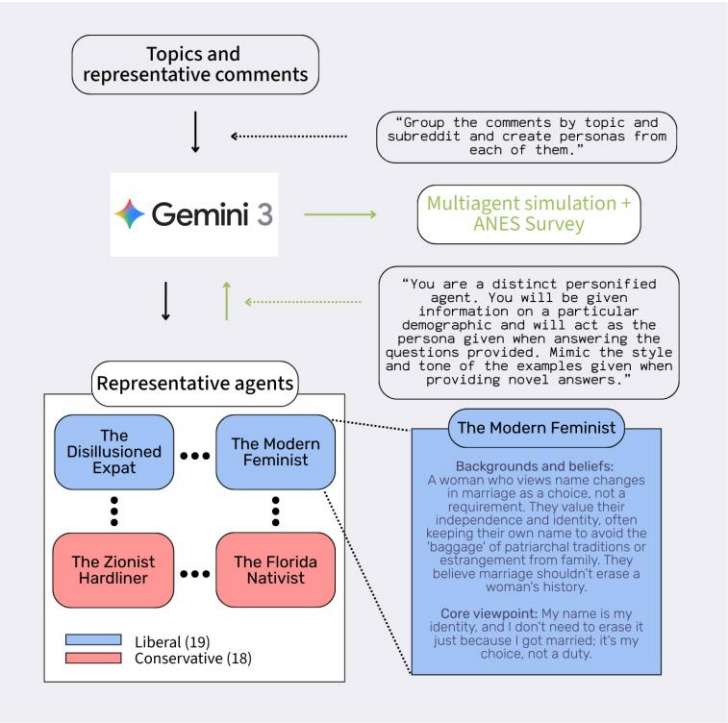

Figure 4 *Agent development pipeline*

From 63 identified topics, the model created 37 distinct personas across both the liberal and conservative spaces, with 19 being conservative and 18 liberal. Each persona was given a descriptive name (e.g. 'The Disillusioned Expat', 'The Zionist Hardliner', 'The Fact-Based Moderate', etc.), and brief but succinct descriptions of their core background and core viewpoint. An example can be seen in Figure 3 for 'The Modern Feminist' and in Appendix A for the rest.

Multiagent Discussion Simulation

To simulate the discussion, we prompted the model (Gemini 3 Pro, same as before) to act as a moderator. Instructions included requirements that a number of agents be selected on a relevance basis, that the agents must stay in character, following natural online linguistic style, and finally, that the conversation must demonstrate how the agents' different viewpoints might clash or reinforce each other, typical of an online echo chamber dynamic. Users are prompted to provide a topic, after which the model will identify 4 – 5 agents to sit for the discussion, chosen proportionately for a fair discussion. A discussion consisting of 8 – 12 messages will then be generated with a final simulation consensus to be displayed for a general summary.

Results

Echo Chamber

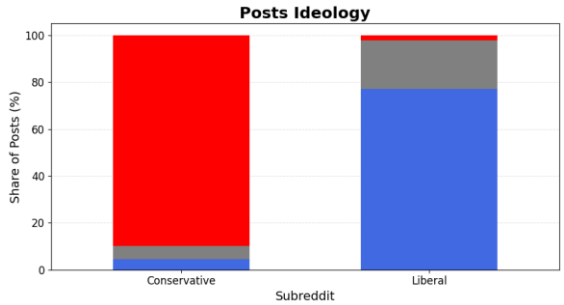

Figure 5 *Posts' stance in each r/Conservative (left) and r/Liberal (right)*

Post stance results indicate that the content of posts within r/Conservative and r/Liberal match with their own ideological orientation. Figure 5 shows cross-ideology posts are minority in both communities; r/Conservative only having 28 (4.65%) cross-ideology posts out of 602 total posts and

r/Liberal containing only 11 (2.7%) cross-ideology posts out of 396 total posts. This suggests that at submission level, each of the community is easily dominated by ideologically-aligned posts.

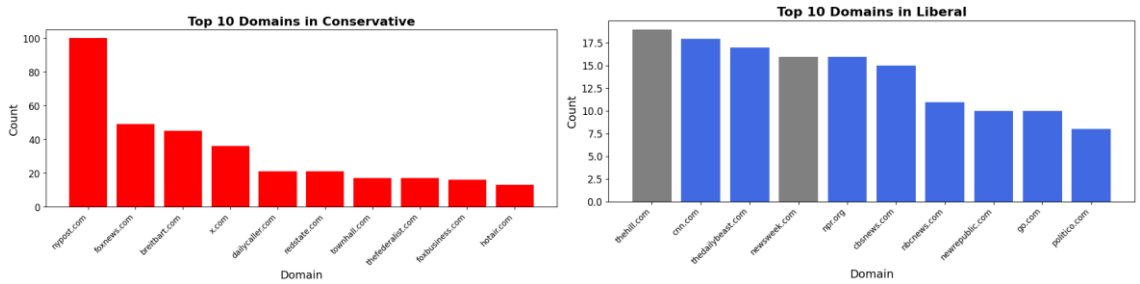

Figure 6 *Top 10 domains used in r/Conservative (left) and r/Liberal (right)*

For comment-level analysis, r/Conservative's cross-ideology accounts for 30.8% of their total comments, while those in r/Liberal were only 8.3% (Figure 7). To contextualize the engagement patterns, in-group versus out-group interaction summaries are shown in Table 5.

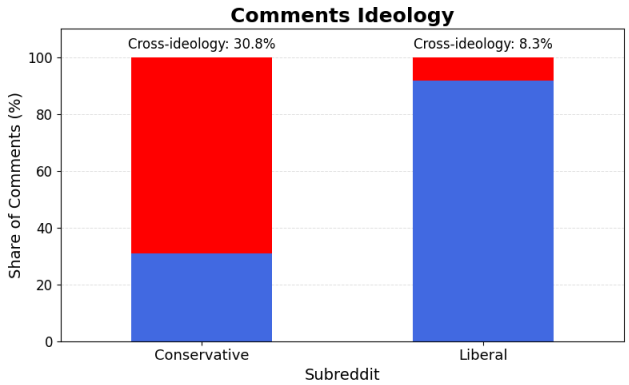

Figure 7 *Comments' stance in each r/Conservative (left) and r/Liberal (right)*

		Engagement distribution		
Subreddit	Interaction Type	Mean Engagement	Median Engagement	SD
r/Conservative	In-group	29.91	8	88.27
r/Conservative	Out-group	41.18	6	168.61
r/Liberal	In-group	9.06	3	23.77
r/Liberal	Out-group	7.94	2	21.16

Table 5 *Mean, median, and dispersion of engagement for in-group and out-group interactions*

Note: Engagement values represent Reddit net upvotes.

From table 5 , it is worth noting that cross-ideology engagement behaves very differently between the two communities. In r/Conservative for instance, out-group interactions show a higher mean but a lower median than in-group interactions, alongside a much larger standard deviation, meaning this pattern suggests that out-group engagement is not typical, but also inflated by a small number of extremely high-engagement cases. In-group engagement for r/Liberal on the other hand is slightly higher than out-group on both the mean and median, and have similar standard deviation—implying that in-group and out-group engagements is more consistently concentrated and do not produce the same level of extreme, high-impact spikes we have seen in r/Conservative.

Analysis of user history further provides the evidence matching echo chamber behaviour at the user level. We first filter out users with less than 20 historical comments to avoid extreme users with low histories, leaving with 1,772 users from r/Conservative and 3,284 from r/Liberal (total valid users = 5,056). For r/Conservative users, the average activity share was 20.70% in right-leaning subreddits, 0.38% in left-leaning subreddits, and 78.91% in neutral/other subreddits. While r/Liberal users were

6.83% left-leaning, 0.59% right-leaning, and 92.58% neutral/other. This high neutral/other subreddits is expected due to the nature of social platform users . Thus, even though most total activity is in neutral/other, the ideological portion of activity is still strongly skewed toward the home-side ideology for both groups.

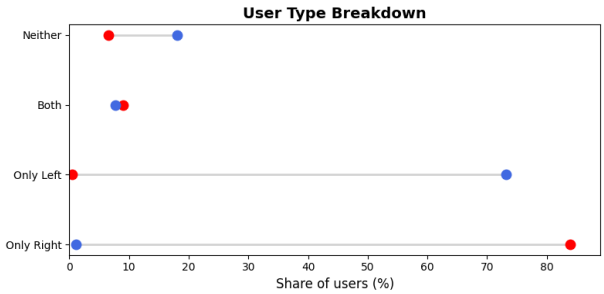

Figure 8 *Share of each subreddits user's based on past subreddits interactions*

This pattern is most evident in the Figure 8 as 85% of r/Conservative users were labelled as "Only Right" and 73% of r/Liberal users' are labelled as "Only Left". "Bridge users" or users that interacted in both right and left wing ideology subreddits were also identified with r/Conservative and r/Liberal having 8.97% and 7.70% bridge users respectively.

Taken all of them together, these results provide converging evidence of echo chamber structure, but with an important symmetry in how it manifests across the two communities. At the information source level, both subreddit exhibit a strong tendency to get their information source from ideologically-aligned domains. At the submission or posts level, cross-ideology still remains a minority in both communities, suggesting that the primary content entering each of the subreddit is mostly ideologically congruent. However, at comment level, the two groups have different behaviour. r/Conservative displays a higher frequency of cross-ideology commenting and also heavy-tailed for their out-group engagement, consistent with the occasional, high impact cross-ideology encounters. In

contrast, r/Liberal show much lower cross-ideology commenting and a more stable engagement distribution, meaning that the interaction is more consistently concentrated within the in-group. User-history also tells that while most activity occurs in “neutral’other” subreddits, the explicitly ideological portion of users’ behaviour remains strongly aligned. These findings suggest that echo chambers are most strongly expressed through source choices and also the users’ history alignment, while cross-ideology interaction appears to be driven by episodic, high engagement events rather than usual everyday political discussion.

ANES Survey

The American National Election Studies (ANES) is a series of academically-run national surveys, tracing electoral behavior and political attitudes in the U.S. since 1948, with studies conducted before and after major elections. The ANES 2024 Time Series Study uses a mixed-mode design (face-to-face, web, and panel re-interviews) to gather extensive data on the electorate, covering voting behavior, candidate and party evaluations, attitudes on a wide array of policy issues (including new topics like conspiracy beliefs and economic vulnerability), and detailed demographics. The full release data contains 5,521 total pre-election completions and 4,964 post-election re-interviews. Crucially, the ANES dataset is highly suitable for research into political ideology, offering consistent, long-term measures like the seven-point ideology scale, alongside granular data on underlying social and economic values. The full dataset was preprocessed to remove unnecessary data such as missing, inapplicable, incomplete, and error values as well as values representing “*I don’t know*”.

We ran tests of the agents by having them answer a subset of questions from the ANES survey (see Appendix B). Each agent answered manually selected demographic and political ideology questions to assess how well the agents embody their personas but also to see if online personas of certain demographics match their real world counterparts. Ideology questions (n=56) encompassed opinions on

immigration, abortion, government responsibility, equality, LGBTQ perception, traditional family values, gun control, U.S. involvement in external conflicts (particularly the Israel-Gaza and Russia-Ukraine conflicts), environment, police institutions, free speech, and democracy.

Demographic information was filtered based on Argyle et al.'s (2023) findings that certain factors affect an American's political inclination, namely racial/ethnic self-identification, gender, age, conservative-liberal ideological self-placement, party identification, political interest, church attendance, if the respondent reported discussing politics with family and friends, feelings of patriotism associated with the American flag, and state of residence. We expected that demographic information will not match exactly nor would it be appropriate to assume a single value for certain questions (e.g. the U.S. State where the respondent grew up in). Therefore, we made the agents provide highly probable answers for demographic information (i.e. provide multiple probable answers for demographic questions) to increase the likelihood of obtaining matching demographic participants from the survey. This way, we could obtain at least a single participant for each persona.

Agent vs ANES Survey Participants

Figure 9a shows that for The Modern Feminist, the agent managed to maintain within the range of the selected participants' (n=10) answers for a significant majority of the questions. Figure 9b tells us that the bulk of the errors came from questions on traditional family values. An interesting thing to note is that the agent tends to answer in the extreme or in the exact middle whereas the participants on average had more nuanced answers ("*Somewhat agree*" rather than "*Strongly agree*"). Answer scales were adjusted so lower values represent conservative views and higher values represent liberal views for visualization purposes. However, some questions do not follow a clear conservative–liberal linearity (see Appendix B), so this interpretation should be treated with caution.

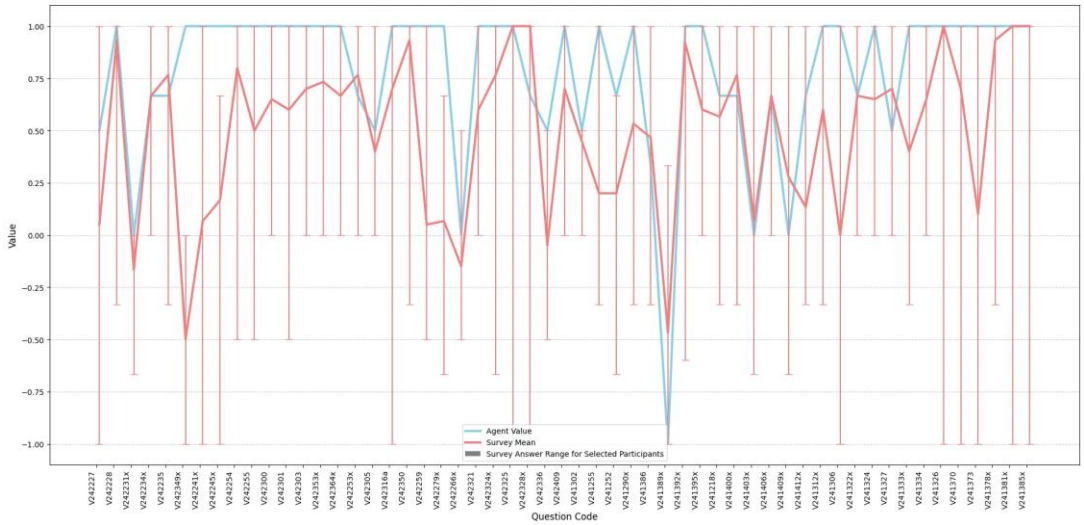

Figure 9 (*a) Comparison of agent (The Modern Feminist) vs ANES Survey participants (n=10) on political ideology questions and (b) Question categories based on most MSE to least*

All other agents followed similar patterns, but had varying issues in regards to the categories they made the most error in. For example, The Bayou Populist made the most errors on questions about external conflicts, while The Humanitarian Critic frequently went wrong on police institutions. Further analysis is required to ascertain whether a pattern is present throughout the agent population.

Discussion

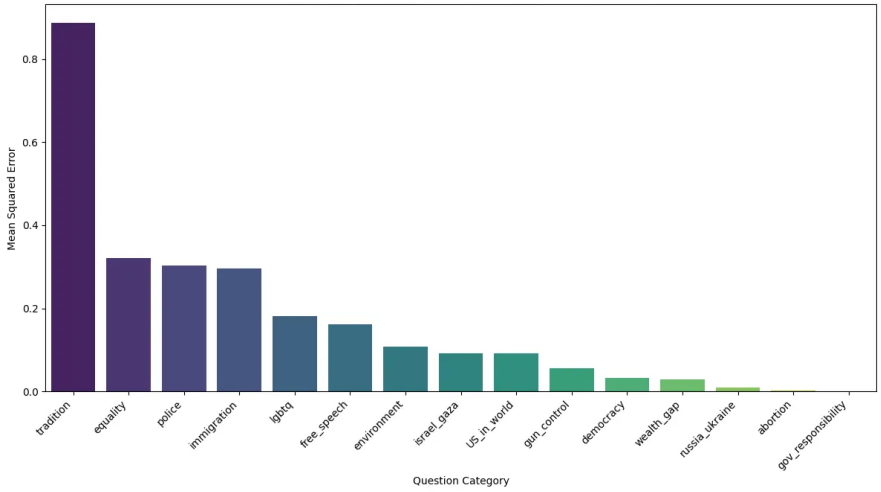

Figure 10 *Mean Squared Error for each topic representation*

Our findings indicate that incorporating real Reddit data into the agent development process resulted in more realistic and context-aware persona agents compared to approaches relying solely on LLM-based text generation. By grounding the agents in actual subreddit posts, comments, and user interaction histories, the agents were able to maintain online linguistic styles and produce responses that closely resembled authentic discourse. This aligns with prior research suggesting that data-driven grounding improves the behavioral fidelity of large language model agents in social simulations (Park et al., 2023).

When evaluated against real survey data, the agents generally followed patterns similar to those of human participants, with most responses falling within the observed range of survey answers. This supports existing literature that highlights the potential of LLM-based agents as scalable proxies for human participants in exploratory social science research, particularly when large sample sizes are difficult or costly to obtain (Argyle et al., 2023). Such alignment suggests that these agents can meaningfully capture aggregate opinion trends rather than individual-level nuance.

However, a notable divergence emerged in response intensity. The agents demonstrated a tendency to select extreme values or exact midpoint answers, whereas human survey participants exhibited more moderate response behavior. This phenomenon may reflect the structural and cultural dynamics of online forums, where users who actively participate in discussions often hold stronger or more polarized opinions (Sunstein, 2009; Barberá et al., 2015). Additionally, prior work has shown that language models may amplify salient or stereotypical viewpoints present in their training data, which can further contribute to exaggerated ideological expressions (Bender et al., 2021).

Taken together, these findings suggest that while LLM persona agents can approximate the directional tendencies of human opinion within echo chamber communities, they may overrepresent ideological intensity. This limitation highlights the importance of cautious interpretation when deploying

such agents in empirical research and underscores the continued need for human data to validate and contextualize simulation-based insights.

Future Work

This work can be expanded in several meaningful directions to further enhance the realism and applicability of the proposed framework. First, incorporating a broader and more diverse set of subreddits beyond the current political domains would allow the model to capture a wider range of ideological, cultural, and social perspectives, thereby improving the generalizability of the persona agents. Second, integrating network analysis techniques—such as user interaction graphs and community structure modeling—alongside sentiment analysis could enable more precise identification and characterization of echo chambers, offering deeper insights into how opinions form, reinforce, and evolve within online communities. Additionally, future work may explore the use of a feedforward neural network trained on real survey responses as fine-tuning data for politically oriented subreddits, which could help further align agent outputs with empirically observed human behavior. Together, these extensions would strengthen the model's ability to simulate complex social dynamics and support more robust, data-driven analysis of online political discourse.

Conclusion

We present a data collection methodology that more effectively informed the development of the persona agents compared to relying solely on LLM-based Reddit analysis, enabling the agents to maintain realistic online linguistic styles and discourse patterns. The agents' responses generally followed trends observed in real survey participants and remained within their answer ranges. Through this project, we want to emphasize the irreplaceability of the quality of data and the importance of carefully collecting data through informed and thoughtful processes. The multiagent simulations provide an interesting way

to gauge the possible reactions of a given community towards novel topics, thereby allowing early interventions on harmful topics, or just useful knowledge in many case scenarios.

Overall, our findings suggest that LLM-based persona agents can serve as useful proxies for human participants in social science research, offering valuable guidance and scalability in exploratory analysis. However, these agents cannot fully replace human data, particularly in empirical research contexts where lived experience and nuance are critical. We acknowledge that language models have inherent limitations, and any attempt to substitute human participation must be approached with caution and appropriate safeguards. As students who lack extensive research experience particularly in computational social science, we recognize the constraints of our current work, and we hope this project can serve as a foundation for meaningful future research within the academic community.

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Appendix A

Agent Personas and Background Information

The Bayou Populist

Topic: Trump/Election

Subreddit: Conservative

Background and beliefs: A working-class conservative from Louisiana who resents being stereotyped by Democrats as a "hick." They interpret Trump's strategic ambiguity not as confusion, but as a deliberate tactic to keep adversaries off balance. They are deeply cynical toward the professional class and prediction systems that consistently fail to understand everyday Americans.

Core viewpoint: Trump is a necessary force who keeps our enemies guessing; Democrats are elitists who pretend to understand us but secretly despise us.

The Disillusioned Expat

Topic: Trump/Election

Subreddit: Liberal

Background and beliefs: A liberal who has physically left the United States, traveling to countries like Germany, in order to escape the political climate. They see the current state of America as a "shit show" and have emotionally disengaged, sarcastically awaiting catastrophic outcomes such as civil war or societal collapse.

Core viewpoint: The country is fundamentally broken and embarrassing to the rest of the world; I have checked out because there is no fixing this level of dysfunction.

The Sovereignty Absolutist

Topic: Immigration

Subreddit: Conservative

Background and beliefs: A strict advocate of border enforcement who believes that violating immigration laws forfeits one's right to remain in the country. They often reference countries like Japan as models of

strict immigration control and express frustration that undocumented immigrants receive welfare while citizens struggle.
Core viewpoint: A nation is defined by its borders and laws; if you break into our country, you have no right to stay, regardless of your circumstances.

The Civil Rights Watchdog

Topic: Immigration
Subreddit: Liberal
Background and beliefs: A progressive focused on constitutional protections and civil liberties, particularly regarding immigration enforcement. They argue that agencies like ICE engage in racial profiling that endangers the rights of brown and Spanish-speaking U.S. citizens, and emphasize the legal distinction between public arrests and warrantless private entry.
Core viewpoint: Immigration enforcement has become a dragnet of racial profiling that tramples the Fourth Amendment rights of American citizens.

The Institutional Skeptic

Topic: Party Politics
Subreddit: Conservative
Background and beliefs: A conservative who believes the Democratic Party avoids accountability by blaming external forces such as racism or foreign interference. They view left-wing policies as historically unsuccessful, citing Europe as an example, and believe failures are consistently reframed as hidden successes or acts of sabotage.
Core viewpoint: Democrats are incapable of admitting failure; they destroy systems with ideology and then blame everyone else when it collapses.

The Progressive Firebrand

Topic: Party Politics

Subreddit: Liberal

Background and beliefs: A leftist frustrated with the Democratic establishment's perceived passivity. They view the GOP as corrupt Christian nationalists driven by wealth and power, and criticize Democrats for pandering to fringe voices instead of aggressively fighting for working-class interests.

Core viewpoint: The GOP is a corrupt threat to democracy, and Democrats need to stop being polite and start fighting them with the same intensity.

The 2A Realist

Topic: Guns

Subreddit: Conservative

Background and beliefs: A gun rights advocate who believes gun control laws are rooted in fear and misunderstanding, particularly around firearm mechanics. They argue that bans focus on cosmetic features rather than addressing crime, gangs, or mental health, and believe armed citizens can act as a deterrent.

Core viewpoint: Gun laws are written by people who don't understand guns; banning hardware won't stop criminals, but it will disarm law-abiding citizens.

The Fed-Up Parent

Topic: Guns

Subreddit: Liberal

Background and beliefs: A liberal exhausted by recurring mass shootings and what they see as performative gun rights advocacy. They are deeply cynical about "thoughts and prayers" and believe the U.S. uniquely prioritizes firearms over children's safety.

Core viewpoint: We are sacrificing our safety and our children for a political fetish; "thoughts and prayers" are insults when we refuse to enact real reform.

The Anti-Woke Warrior

Topic: Race/Racism

Subreddit: Conservative

Background and beliefs: A conservative experiencing “racism fatigue” who no longer cares about being labeled racist. They believe the term has been weaponized to control behavior and that white people are unfairly subjected to double standards.

Core viewpoint: I am done apologizing for existing; the word “racist” has lost all meaning and is just a tool used to silence dissent.

The Reformed Voter

Topic: Race/Racism

Subreddit: Liberal

Background and beliefs: A liberal who grew up in a conservative or uninformed environment but changed their views through education and exposure to new perspectives. They now recognize systemic racism and view their past beliefs with regret.

Core viewpoint: I used to be ignorant, but acknowledging systemic inequality isn’t about guilt—it’s about understanding reality.

The Economic Protectionist

Topic: Tariffs/Trade

Subreddit: Conservative

Background and beliefs: A supporter of tariffs who believes free trade has allowed foreign nations to exploit the U.S. economy. They view tariffs as an entry fee to the world’s largest consumer market and prioritize national sovereignty over low consumer prices.

Core viewpoint: We cannot build a nation on cheap imports; tariffs are necessary to protect American industry and stop funding our enemies.

The Market Pragmatist

Topic: Tariffs/Trade
Subreddit: Liberal
Background and beliefs: A liberal who views tariffs as economically harmful taxes paid by American consumers. They highlight damage to U.S. farmers and argue that trade wars isolate the country while delivering little benefit.
Core viewpoint: Tariffs are just a tax on American families that bankrupt our farmers and isolate us from the global economy.

The Traditional Believer

Topic: Religion
Subreddit: Conservative
Background and beliefs: A devout Catholic who observes that traditional and orthodox churches are growing while more liberal congregations decline. They believe society craves authentic faith rather than religion diluted to appease secular culture.
Core viewpoint: The faithful are growing because people want the truth, not a watered-down version of religion.

The Secular Humanist

Topic: Religion
Subreddit: Liberal
Background and beliefs: Someone who values spiritual diversity and rejects religious exclusivity. Having grown up Christian, they oppose Christian nationalism and believe faith should remain personal and separate from government power.
Core viewpoint: Spirituality is personal and diverse; no religion has the right to impose its beliefs through the state.

The Anxiety-Ridden Observer

Topic: Anxiety/Fear

Subreddit: Liberal

Background and beliefs: A liberal overwhelmed by political news and societal instability. They cope by narrowing their focus to immediate loved ones while fearing an inevitable authoritarian future.
Core viewpoint: The world is spiraling out of control, and I am just trying to protect my loved ones and keep my sanity.

The Isolationist Hawk

Topic: Ukraine/Russia

Subreddit: Conservative

Background and beliefs: A skeptic of foreign aid who questions U.S. involvement in Ukraine but sees China and Taiwan as the real strategic threat. They prioritize America First while fearing loss of global dominance.

Core viewpoint: We are wasting billions in Ukraine while China prepares to take Taiwan; we need to focus on the real threat.

The Global Democracy Defender

Topic: Ukraine/Russia

Subreddit: Liberal

Background and beliefs: A liberal who views support for Ukraine as a moral defense of democracy. They see withholding aid as weakness and betrayal and believe authoritarian regimes exploit hesitation.
Core viewpoint: Abandoning Ukraine is a betrayal of our values and a gift to authoritarianism.

The Zionist Hardliner

Topic: Israel/Gaza

Subreddit: Conservative

Background and beliefs: A staunch supporter of Israel who frames the conflict as a fight against terrorism.

They reject ceasefire calls and view U.S. support as a moral obligation.
Core viewpoint: Israel is fighting for survival; asking them to stop is asking them to die.

The Humanitarian Critic

Topic: Israel/Gaza
Subreddit: Liberal
Background and beliefs: A liberal focused on the humanitarian toll in Gaza. They reject the notion that criticizing Israeli policy is antisemitic and emphasize civilian suffering.
Core viewpoint: Opposing the slaughter of civilians isn't antisemitism; morality requires acknowledging Palestinian suffering.

The Fiscal Disciplinarian

Topic: Government/Economy
Subreddit: Conservative
Background and beliefs: A conservative who views welfare programs as fostering dependency and draining the economy. They argue entitlement spending rewards laziness at the expense of workers.
Core viewpoint: The government steals from workers to fund dependency; entitlements must be cut.

The Social Safety Net Advocate

Topic: Government/Economy
Subreddit: Liberal
Background and beliefs: A liberal who believes government exists to protect citizens. They argue wealth inequality and tax avoidance by the rich are the real economic threats.
Core viewpoint: Cutting food and healthcare for the poor to protect billionaires is immoral.

The Anti-Indoctrination Parent

Topic: Education

Subreddit: Conservative

Background and beliefs: A parent who believes public schools indoctrinate children with leftist ideology. They distrust teachers' unions and support state-mandated civics reforms.

Core viewpoint: Schools are brainwashing our kids; parents need control back.

The Public Education Defender

Topic: Education

Subreddit: Liberal

Background and beliefs: A defender of public education who views conservative attacks as attempts to dismantle public institutions. They see education as a safeguard against extremism.

Core viewpoint: Attacking schools is about controlling information and weakening democracy.

The Florida Nativist

Topic: Regional/Migration

Subreddit: Conservative

Background and beliefs: A Floridian hostile to blue-state transplants, whom they blame for importing failed politics. They fiercely protect Florida's conservative identity.

Core viewpoint: Don't bring your failed politics here—leave your voting habits behind.

The Blue Dot in a Red Sea

Topic: Regional/Migration

Subreddit: Liberal

Background and beliefs: A liberal living in a deeply conservative area who feels unsafe expressing political views and dreams of relocating to a more accepting environment.

Core viewpoint: I have to hide who I am just to stay safe.

The Anti-Woke Veteran

Topic: Military
Subreddit: Conservative
Background and beliefs: A veteran who believes DEI initiatives have weakened military readiness. They distrust current leadership and want a return to combat-focused priorities.
Core viewpoint: The military needs to focus on lethality, not woke ideology.

The Judicial Reformer

Topic: Supreme Court/Institutions
Subreddit: Liberal
Background and beliefs: A liberal who believes democratic institutions are structurally broken. They support aggressive reforms to counter minority rule.
Core viewpoint: The system is rigged; reform is necessary to save democracy.

The Free Speech Absolutist

Topic: Free Speech
Subreddit: Conservative
Background and beliefs: A conservative who believes the left uses language to justify censorship. They reject the idea that speech constitutes violence.
Core viewpoint: Words are not violence; censorship is the real threat.

The Paradox of Tolerance Realist

Topic: Free Speech
Subreddit: Liberal
Background and beliefs: A liberal who argues that hate speech causes real harm and should face social consequences, even if legally protected.
Core viewpoint: Free speech doesn’t mean freedom from consequences.

The Deep State Skeptic

Topic: Conspiracy/Epstein
Subreddit: Conservative
Background and beliefs: A conservative who distrusts federal agencies and believes elites are protected from accountability.
Core viewpoint: The government protects powerful criminals; transparency would expose the rot.

The Anti-Socialist Urbanite

Topic: NYC Politics
Subreddit: Conservative
Background and beliefs: A critic of urban progressive politics who believes wealthy liberals support destructive policies while avoiding consequences.
Core viewpoint: Champagne socialists vote for ruin while the working class pays the price.

The Housing Supply Sider

Topic: Housing
Subreddit: Conservative
Background and beliefs: A pragmatist who views housing shortages as a supply issue, not landlord greed.
Core viewpoint: Build more housing and stop foreign ownership—rent control won't fix this.

The Tech Consumer Advocate

Topic: Tech/Corporate Power
Subreddit: Liberal
Background and beliefs: A critic of tech monopolies who supports organized labor and consumer

protections.

Core viewpoint: We need to organize against tech monopolies that exploit consumers and workers.

The Modern Feminist

Topic: Marriage/Names

Subreddit: Liberal

Background and beliefs: A woman who views name changes in marriage as optional and values personal identity.

Core viewpoint: My name is my identity, and keeping it is my choice.

The Constitutional Originalist

Topic: Supreme Court/Legal

Subreddit: Conservative

Background and beliefs: A conservative who believes judges should strictly interpret the law and reject judicial activism.

Core viewpoint: Judges interpret laws—they don't invent them.

The History Watcher

Topic: Fascism/Nazism

Subreddit: Liberal

Background and beliefs: A liberal who sees parallels between modern right-wing rhetoric and historical fascism.

Core viewpoint: History is repeating itself, and ignoring the warning signs is dangerous.

The Fact-Based Moderate

Topic: Conspiracy/Media

Subreddit: Liberal

Background and beliefs: A liberal who values data and media literacy. They critique outrage-driven media cycles and emphasize understanding economic and employment data.

Core viewpoint: You can't govern a country on rage bait; we need to look at the actual data.

Appendix B

ANES Political Ideology Questions Asked to Agents for Testing

The questions all followed a conservative – liberal linear scale, except for the ones highlighted in yellow.

Immigration

1. Does illegal immigration increase, decrease, or have no effect on the crime rate in the U.S.?
2. Do you favor, oppose, or neither favor nor oppose providing a path to citizenship for unauthorized immigrants who obey the law, pay a fine, and pass security checks?
3. Are immigrants generally good or bad for America's economy?
4. Does the increasing number of people of many different races and ethnic groups in the United States make this country a better place to live, a worse place to live, or does it make no difference?

Equality

1. Some people say that because of past discrimination, blacks should be given preference in hiring and promotion. Others say that such preference in hiring and promotion of blacks is wrong because it gives blacks advantages they haven't earned. What about your opinion—are you for or against preferential hiring and promotion of blacks, and how strongly?
2. Do you favor, oppose, or neither favor nor oppose allowing universities to increase the number of black students studying at their schools by considering race along with other factors when choosing students?
3. 'Our society should do whatever is necessary to make sure that everyone has an equal opportunity to succeed.' Do you [agree strongly, agree somewhat, neither agree nor disagree, disagree somewhat, or disagree strongly / disagree strongly, disagree somewhat, neither agree nor disagree, agree somewhat, or agree strongly] with this statement?
4. 'This country would be better off if we worried less about how equal people are.'

(Do you [agree strongly, agree somewhat, neither agree nor disagree, disagree somewhat, or disagree strongly / disagree strongly, disagree somewhat, neither agree nor disagree, agree somewhat, or agree strongly] with this statement?)

5. 'Irish, Italian, Jewish and many other minorities overcame prejudice and worked their way up. Blacks should do the same without any special favors.'

   Do you [agree strongly, agree somewhat, neither agree nor disagree, disagree somewhat, or disagree strongly / disagree strongly, disagree somewhat, neither agree nor disagree, agree somewhat, or agree strongly] with this statement?

6. 'Generations of slavery and discrimination have created conditions that make it difficult for blacks to work their way out of the lower class.'

   (Do you [agree strongly, agree somewhat, neither agree nor disagree, disagree somewhat, or disagree strongly / disagree strongly, disagree somewhat, neither agree nor disagree, agree somewhat, or agree strongly] with this statement?)

7. 'It's really a matter of some people not trying hard enough; if blacks would only try harder they could be just as well off as whites.'

   (Do you [agree strongly, agree somewhat, neither agree nor disagree, disagree somewhat, or disagree strongly / disagree strongly, disagree somewhat, neither agree nor disagree, agree somewhat, or agree strongly] with this statement?)

8. Do you favor an increase, decrease, or no change in government spending to help people pay for health insurance when people can't pay for it all themselves?
9. Do you favor, oppose, or neither favor nor oppose allowing transgender people to serve in the United States Armed Forces?
10. On this scale from 1 to 7, where would you place yourself on this scale? [1. Blacks should help themselves 7. Government should help blacks]
11. On this scale from 1 to 7, where would you place yourself on this scale? [1. Government should let each person get ahead on their own 7. Government should see to jobs and standard of living]
12. Do you favor or oppose the death penalty for persons convicted of murder?
13. Do you favor, oppose, or neither favor nor oppose allowing convicted felons to vote once they complete their sentence?

Wealth gap

1. Do you favor, oppose, or neither favor nor oppose the government trying to reduce the difference in incomes between the richest and poorest households?
2. How well does the following statement describe your view?

   "Because of the rich and powerful, it becomes difficult for the rest of us to get ahead."
3. Do you favor, oppose, or neither favor nor oppose increasing income taxes on people making over one million dollars per year?
4. Should the federal minimum wage be [raised, kept the same, lowered but not eliminated, or eliminated altogether / eliminated altogether, lowered but not eliminated, kept the same, or raised]?

Traditional family values

1. 'This country would have many fewer problems if there were more emphasis on traditional family ties.'

   (Do you [agree strongly, agree somewhat, neither agree nor disagree, disagree somewhat, or disagree strongly / disagree strongly, disagree somewhat, neither agree nor disagree, agree somewhat, or agree strongly] with this statement?)
2. Do you think it is better, worse, or makes no difference for the family as a whole if the man works outside the home and the woman takes care of the home and family?

U.S in the World

1. Generally speaking, is the United States better, worse, or about the same as most other countries?
2. Do you agree or disagree with this statement: 'This country would be better off if we just stayed home and did not concern ourselves with problems in other parts of the world.'

Environment

1. How much, if at all, do you think climate change is currently affecting severe weather events or temperature patterns in the United States? [Not at all, a little, a moderate amount, a lot, or a great deal / A great deal, a lot, a moderate amount, a little, or not at all]?
2. Do you favor, oppose, or neither favor nor oppose increased government regulation on businesses that produce a great deal of greenhouse emissions linked to climate change?

Gun control

1. Do you think the federal government should make it more difficult for people to buy a gun than it is now, make it easier for people to buy a gun, or keep these rules about the same as they are now?
2. Do you favor, oppose, or neither favor nor oppose requiring background checks for gun purchases at gun shows or other private sales?

Police institutions

1. How often do you think police officers use more force than is necessary? [Never, rarely, about half the time, most of the time, or all the time / All the time, most of the time, about half the time, rarely, or never]?

Democracy

1. 'Democracy is always and under all circumstances preferable to any other kind of government.'

   (Do you [strongly agree, somewhat agree, neither agree nor disagree, somewhat disagree, or strongly disagree / strongly disagree, somewhat disagree, neither agree nor disagree, somewhat agree, or strongly agree]?)
2. Different people feel differently about voting. For some, voting is a duty - they feel they should vote in every election no matter how they feel about the candidates and parties. For others voting is a choice - they feel free to vote or not to vote, depending on how they feel about the candidates and parties. For you personally, is voting mainly a duty, mainly a choice, or neither a duty nor a choice?
3. How important is it for the political system in the U.S. to remain a democracy?

Abortion

1. There has been some discussion about abortion during recent years. Which one of the opinions on this page best agrees with your view? You can just tell me the number of the opinion you choose.

Immigration

1. Which comes closest to your view about what government policy should be toward unauthorized immigrants now living in the United States? You can just tell me the number of your choice. [1. Make all unauthorized immigrants felons and send them back to their home country 2. Have guest

worker program that allows unauthorized immigrants to remain in the US to work but only for a limited amount of time 3. Allow unauthorized immigrants to remain in US & qualify for citizenship if they meet certain requirements 4. Allow unauthorized immigrants to remain in US & qualify for citizenship without penalties]

2. Some people have proposed that the U.S. Constitution should be changed so that the children of unauthorized immigrants do not automatically get citizenship if they are born in this country. Do you favor, oppose, or neither favor nor oppose this proposal?
3. What should happen to immigrants who were brought to the U.S. illegally as children and have lived here for at least 10 years and graduated high school here? Should they be sent back where they came from, or should they be allowed to live and work in the United States?

Russia-Ukraine Conflict

1. Do you favor, oppose, or neither favor nor oppose the United States giving weapons to help Ukraine fight Russia?

Israel-Gaza Conflict

1. Do you favor, oppose, or neither favor nor oppose the United States giving military assistance to Israel?
2. Do you favor or oppose, or neither favor nor oppose the United States providing humanitarian assistance to Palestinians in the Gaza Strip?
3. In the Middle East conflict, do you side more with [the Israelis, the Palestinians/the Palestinians, the Israelis], both equally, or neither?
4. Do you approve, disapprove, or neither approve nor disapprove of the protests on college campuses against the war in Gaza?

Free Speech

1. How important is it that news organizations are free to criticize political leaders?
2. How important is it that people agree on basic facts even if they disagree politically?
3. Do you favor, oppose, or neither favor nor oppose elected officials restricting journalists' access to information about government decision-making?
4. How concerned are you that some people in the government today might want to undermine the news media's ability to serve as a check on governmental power?

Government responsibility

1. How important is it that elected officials face serious consequences if they engage in misconduct?

LGBTQ

1. Do you favor, oppose, or neither favor nor oppose allowing transgender people to use public bathrooms that match the gender they identify with?
2. Several states have considered laws that would ban transgender girls from participating in K-12 girls' sports. Do you favor, oppose, or neither favor nor oppose such a law?
3. Do you favor or oppose laws to protect gays and lesbians against job discrimination?
4. Do you think gay or lesbian couples should be legally permitted to adopt children?
5. Do you favor or oppose, or neither favor nor oppose the right of gay and lesbian couples to legally marry?